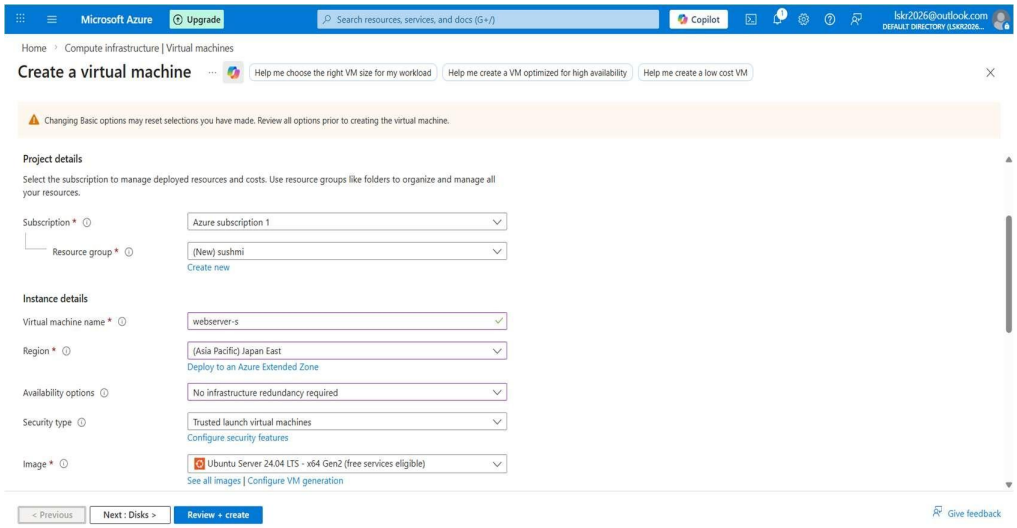
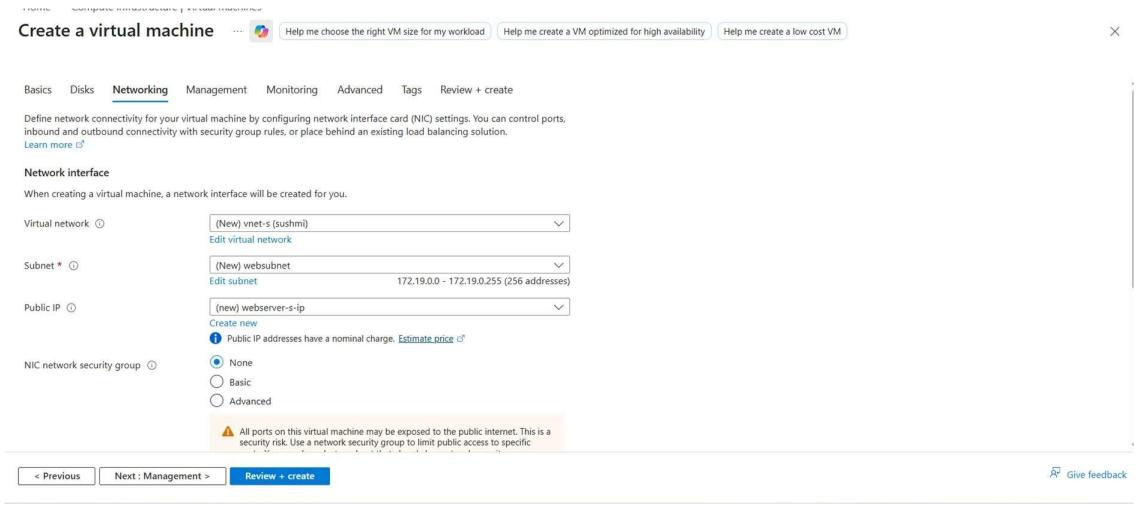
APPLICATION SECURITY GROUP

- TO CREATE A 2 VIRTUAL MACHINES
- WEB-SERVER AND APP-SERVER WITH OUT NSG

WEB-SERVER VM





Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

Custom data and cloud init

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. Learn more about custom data for VMs

Custom data

```
#!/bin/bash
sudo su
apt update
apt install nginx -y
```

Custom data on the selected image will be processed by cloud-init.
Learn more about custom data for VMs

User data

Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual machine. Don't use user data for storing your secrets or passwords. Learn more about user data for VMs

Enable user data

Performance (NVMe)

Enable capabilities to enhance the performance of your resources.

< Previous | Next : Tags > | Review + create

Give feedback

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260130141955 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete | Cancel | Redeploy | Download | Refresh

Deployment succeeded

Deployment 'CreateVm-canonical.ubuntu-24_04-lts-server-20260130141955' to resource group 'sushmi' was successful.

Go to resource | Pin to dashboard

Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2...
Subscription: Azure subscription 1
Resource group: sushmi

Start time: 1/30/2026, 2:26:57 PM
Correlation ID: bf08e140-b9c3-4659-9e0f-34fb5d2ff92d

Deployment details

Next steps

Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource | Create another VM

Give feedback

Tell us about your experience with deployment

Cost Management

Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners.

APP SERVER VM

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network: vnet-s (sushmi)
Edit virtual network

Subnet *: (New) app-subnet
Edit subnet — 172.19.1.0 - 172.19.1.255 (256 addresses)

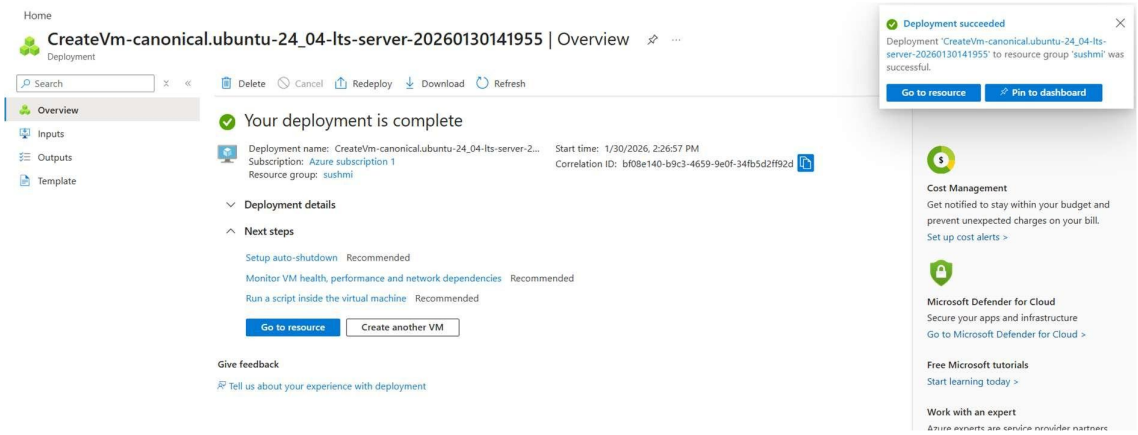
Public IP: (new) app-sever-s-ip
Create new
Public IP addresses have a nominal charge. Estimate price

NIC network security group:
- None
- Basic
- Advanced

All ports on this virtual machine may be exposed to the public internet. This is a security risk. Use a network security group to limit public access to specific ports. You can also select a subnet that already has network security groups defined or remove the public IP address.

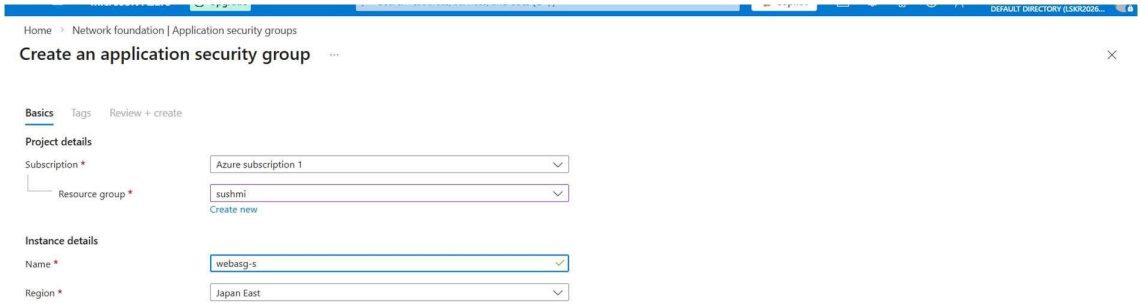
< Previous | Next : Management > | Review + create

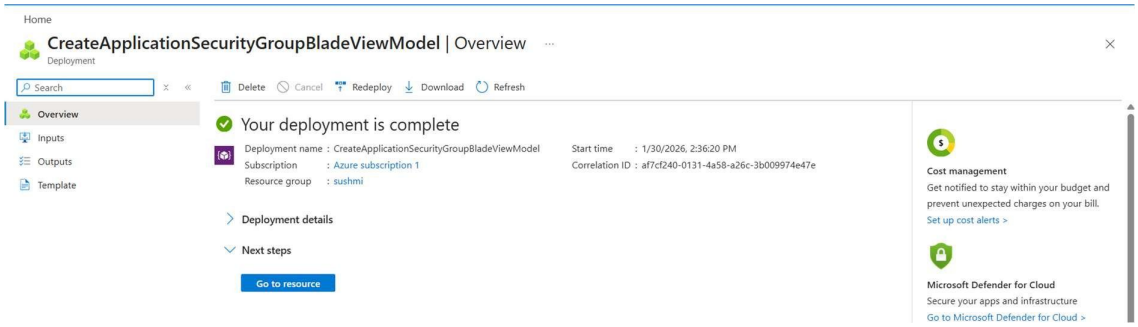
Give feedback

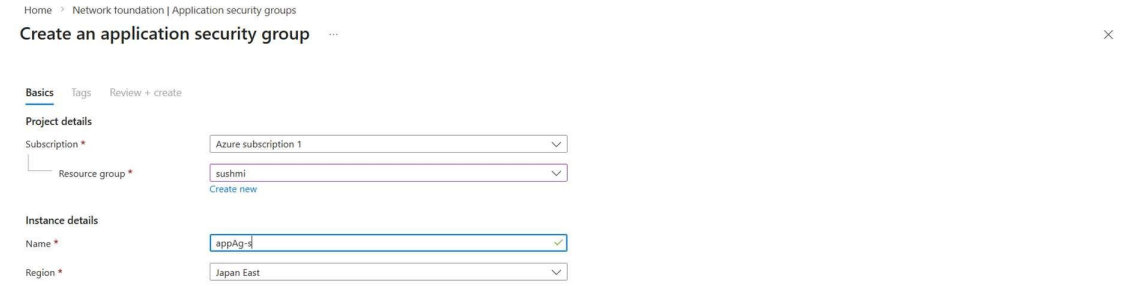


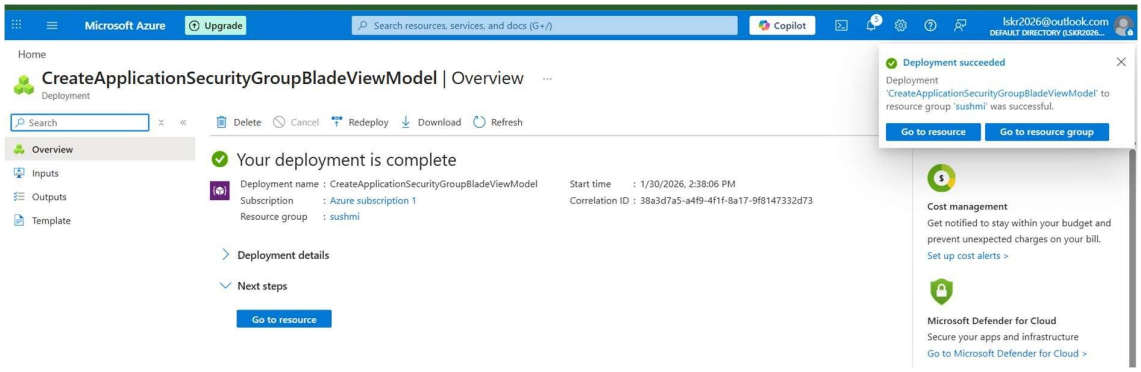
NOW WE CAN CREATE A APPLICATION SECURITY GROUP

WEBASG AND APPASG



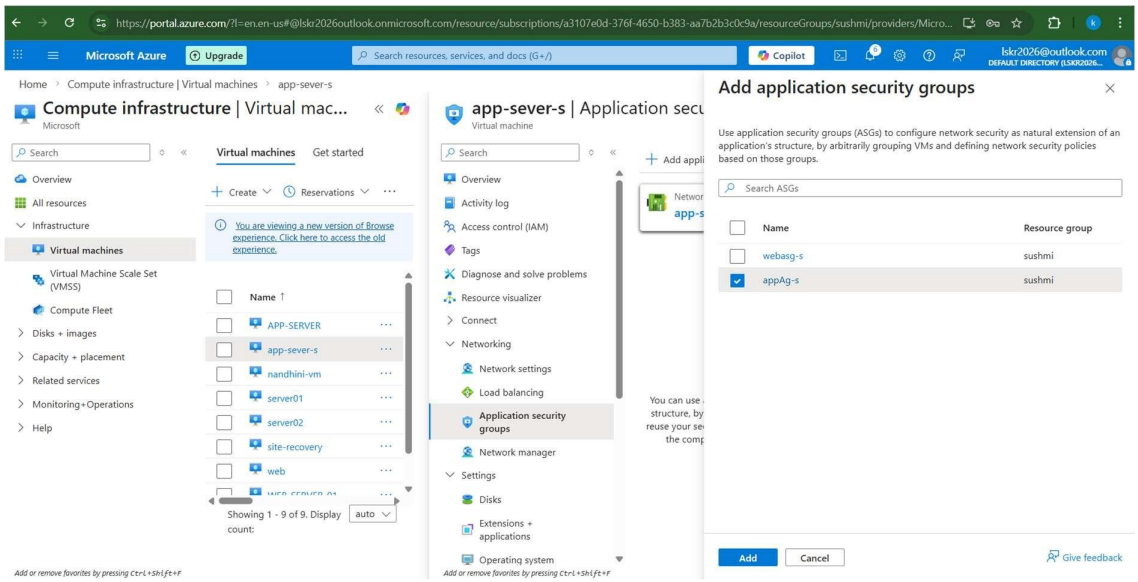


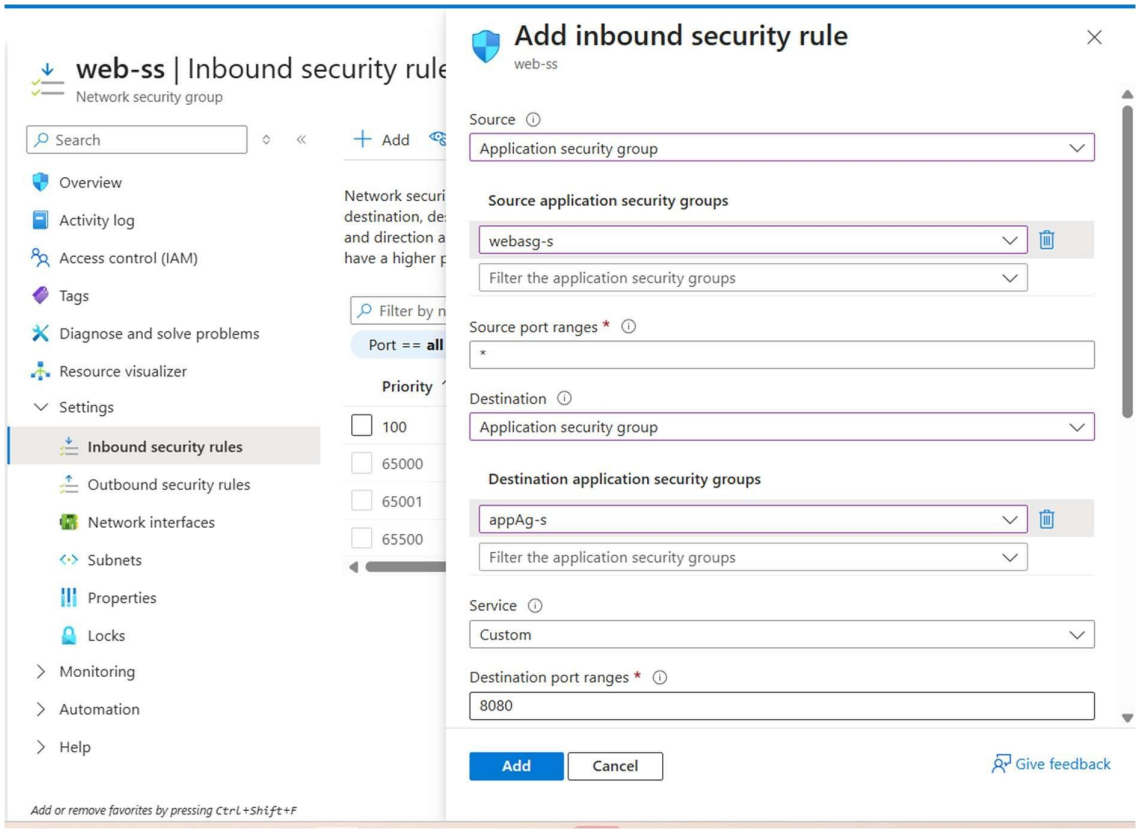


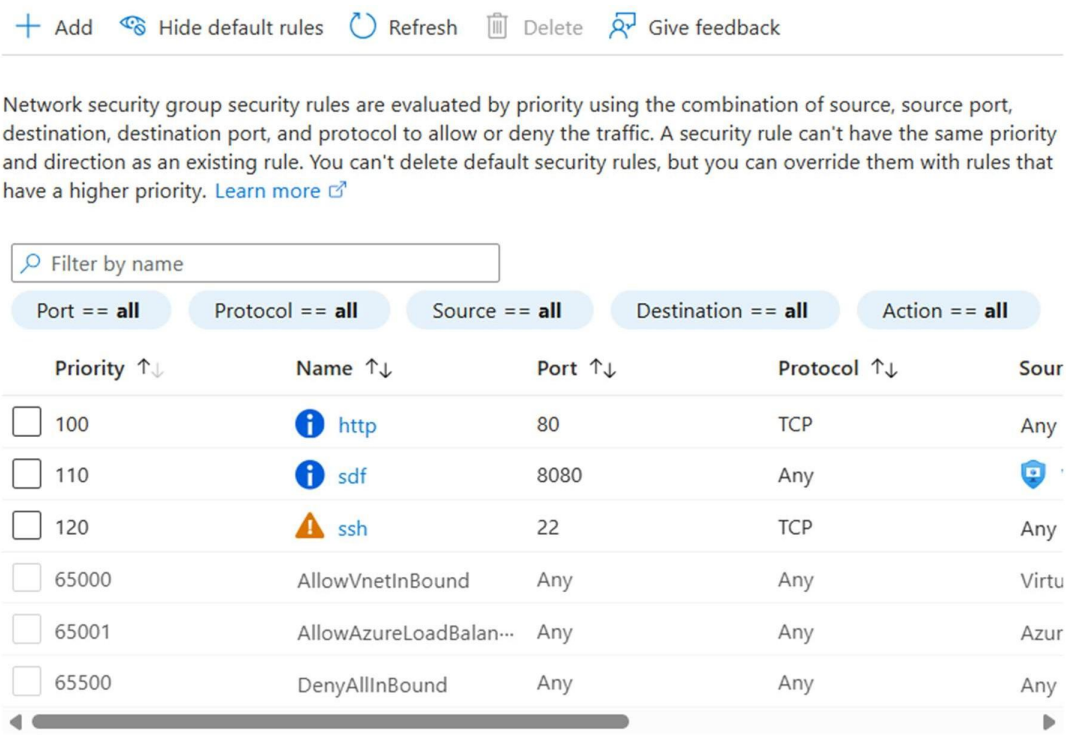


ADD WEBASG TO WEBSERVER

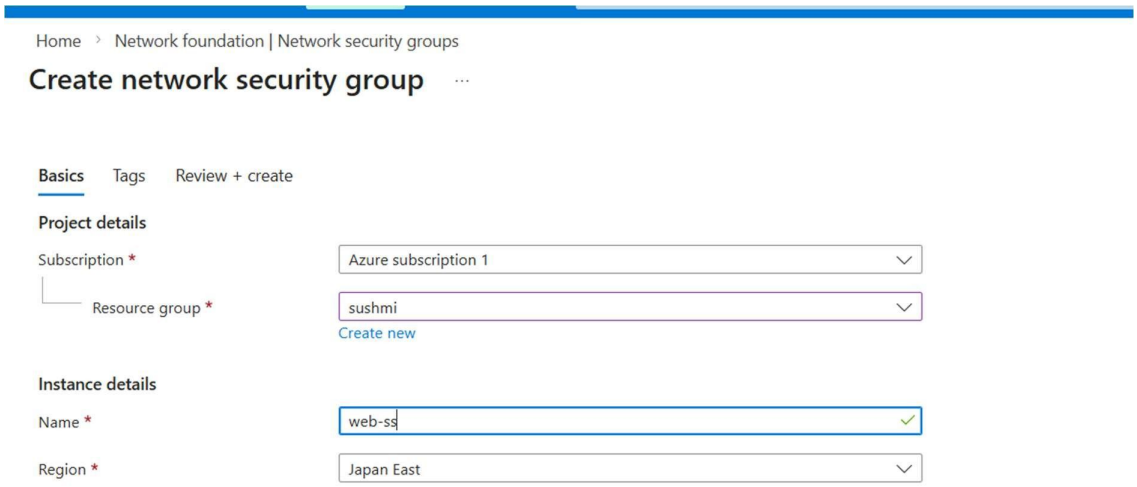
APP-ASG TO APP-SERVER



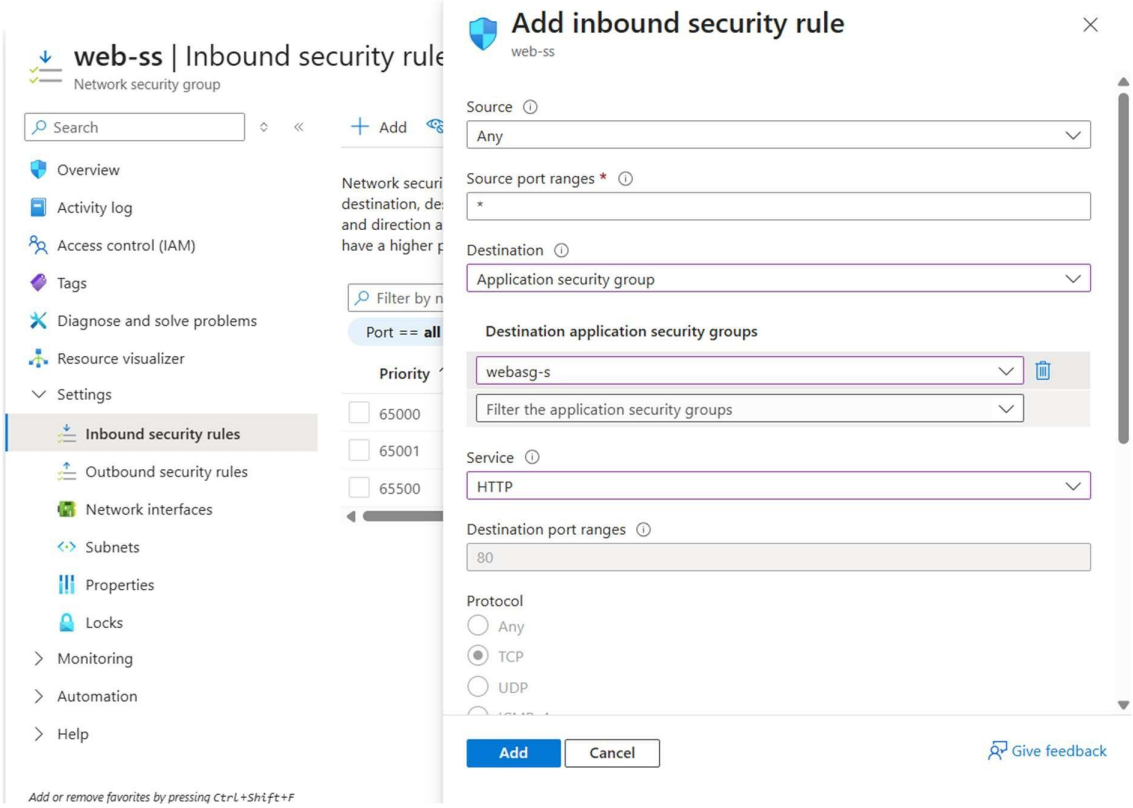

web-ss | Inbound security rule
Network security group
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties
Locks
Monitoring
Automation
Help
Add or remove favorites by pressing Ctrl+Shift+F
Add
Filter by n
Port == all
Priority
100
65000
65001
65500
Add inbound security rule
web-ss
Source
Application security group
Source application security groups
webasg-s
Filter the application security groups
Source port ranges *
*
Destination
Application security group
Destination application security groups
appAg-s
Filter the application security groups
Service
Custom
Destination port ranges *
8080
Add
Cancel
Give feedback

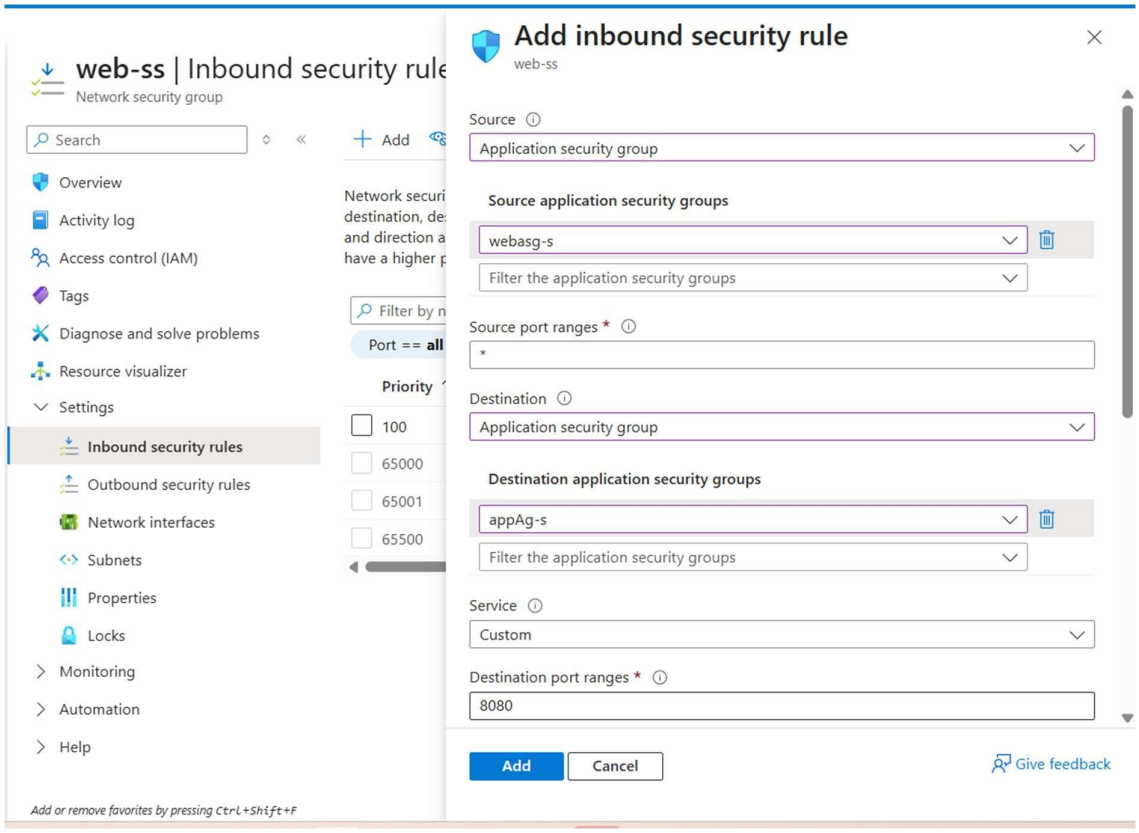

Add
Hide default rules
Refresh
Delete
Give feedback
Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. Learn more
Filter by name
Port == all
Protocol == all
Source == all
Destination == all
Action == all
Priority
Name
Port
Protocol
Sour
100 http 80 TCP Any
110 sdf 8080 Any
120 ssh 22 TCP Any
65000 AllowVnetInBound Any Any Virtu
65001 AllowAzureLoadBalan··· Any Any Azur
65500 DenyAllInBound Any Any Any

NOW WE CAN CREATE A NETWORK SECURITY GROUP

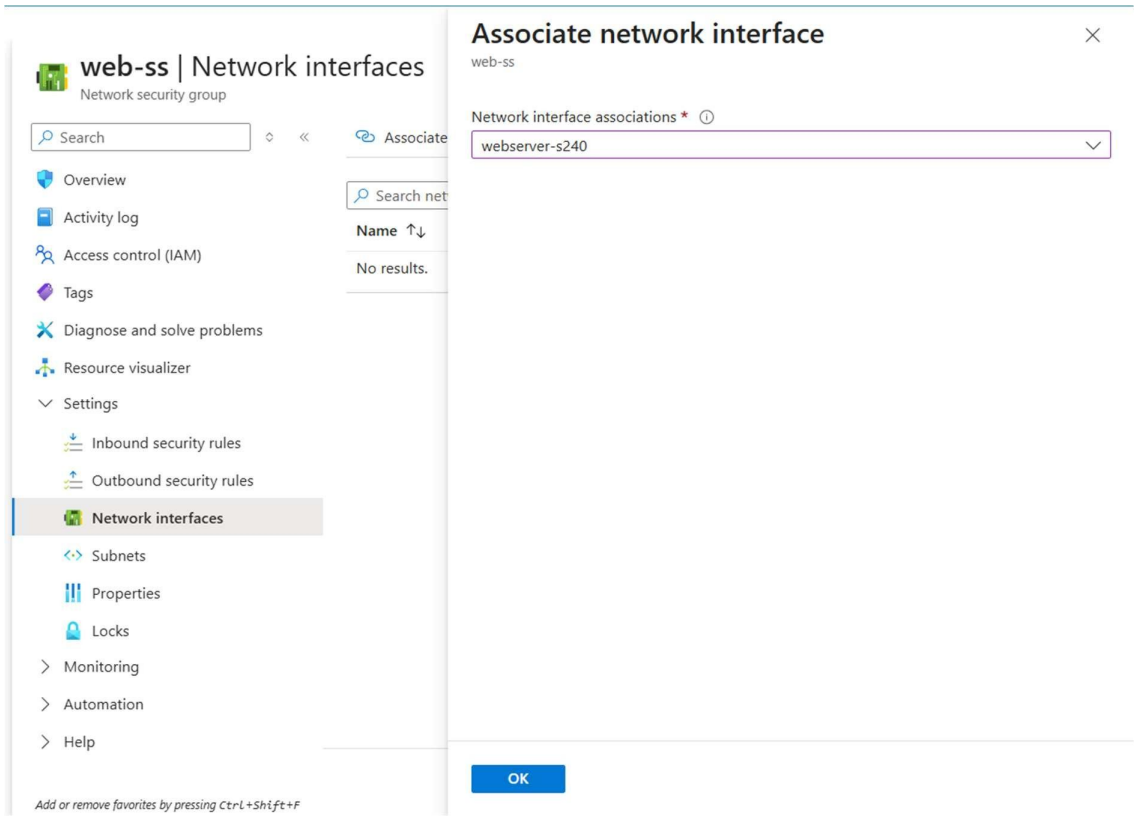


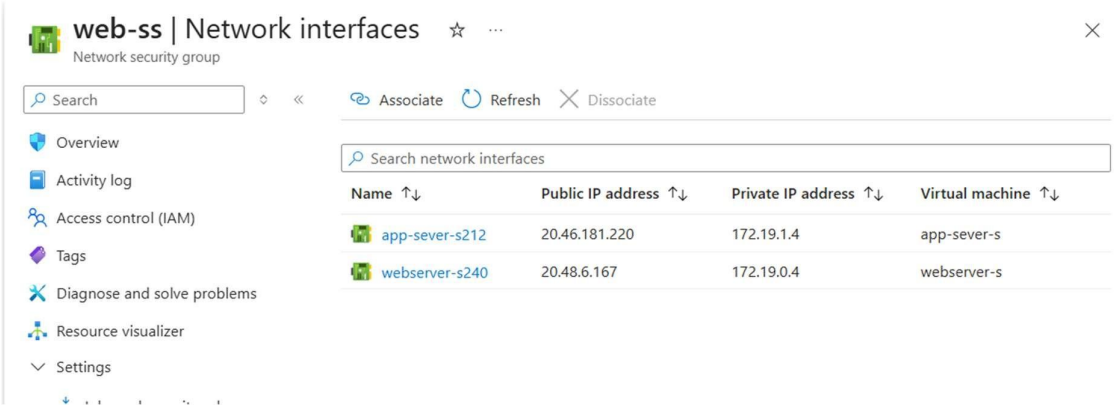
ADD THE INBOUND RULES



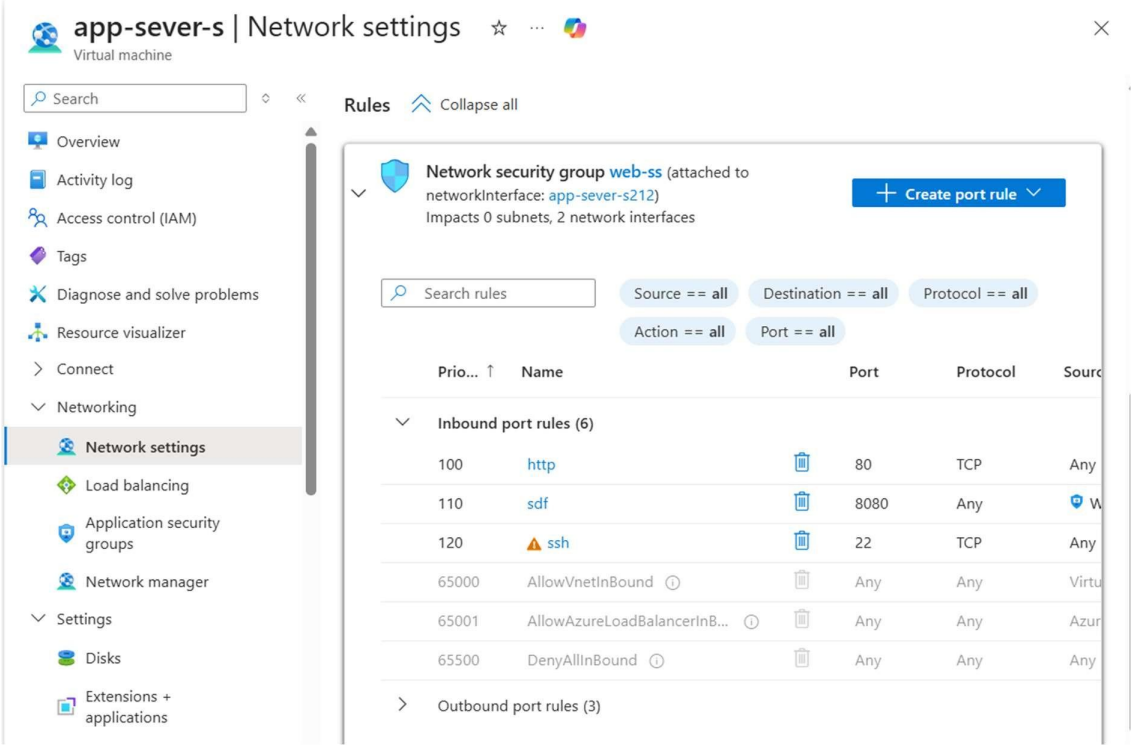


ASSOCIATE THE NETWORK INTERFACE





NOW WE CAN SEE THE RULES IN BOTH WEB-SERVER AND APP-SERVER



webserver-s | Network settings
Virtual machine

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect
- Networking
  - Network settings
  - Load balancing
  - Application security groups
  - Network manager
- Settings
  - Disks
  - Extensions +

Network security group web-ss (attached to networkInterface: webserver-s240)
Impacts 0 subnets, 2 network interfaces

Create port rule

Search rules | Source == all | Destination == all | Protocol == all | Action == all | Port == all

Prio...	Name	Port	Protocol	Sourc
Inbound port rules (6)				
100	http	80	TCP	Any
110	sdf	8080	Any	W
120	ssh	22	TCP	Any
65000	AllowVnetInBound	Any	Any	Virtu
65001	AllowAzureLoadBalancerInB...	Any	Any	Azur
65500	DenyAllInBound	Any	Any	Any
Outbound port rules (3)				

Folder shared with you: 'Vcube... | CLOUD-RECORDINGS - Google | Welcome to nginx!

Not secure 20.48.6.167

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

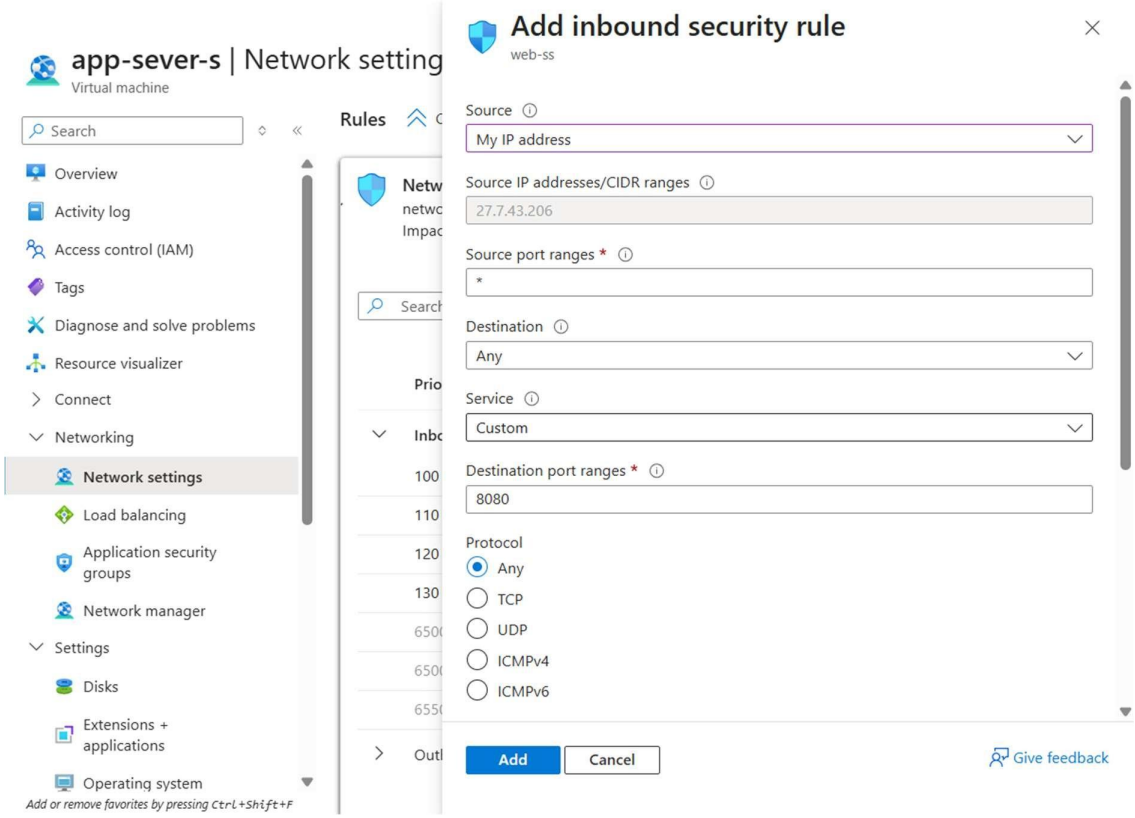
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
apache-tomcat-11.0.18/bin/ciphers.sh
apache-tomcat-11.0.18/bin/configtest.sh
apache-tomcat-11.0.18/bin/daemon.sh
apache-tomcat-11.0.18/bin/digest.sh
apache-tomcat-11.0.18/bin/makebase.sh
apache-tomcat-11.0.18/bin/migrate.sh
apache-tomcat-11.0.18/bin/setclasspath.sh
apache-tomcat-11.0.18/bin/shutdown.sh
apache-tomcat-11.0.18/bin/startup.sh
apache-tomcat-11.0.18/bin/tool-wrapper.sh
apache-tomcat-11.0.18/bin/version.sh
root@app-sever-s:/home/sushmi# ls
apache-tomcat-11.0.18  apache-tomcat-11.0.18.tar.gz
root@app-sever-s:/home/sushmi# mv apache-tomcat-11.0.18 tomcat
root@app-sever-s:/home/sushmi# cd tomcat/bin
root@app-sever-s:/home/sushmi/tomcat/bin# ./startup.sh
Using CATALINA_BASE:   /home/sushmi/tomcat
Using CATALINA_HOME:   /home/sushmi/tomcat
Using CATALINA_TMPDIR: /home/sushmi/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/sushmi/tomcat/bin/bootstrap.jar:/home/sushmi/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@app-sever-s:/home/sushmi/tomcat/bin# telnet 172.19.0.4 8080
Trying 172.19.0.4...
```

```
telnet: Unable to connect to remote host: Connection refused
root@webserver-s:/home/sushmi# telnet 172.19.1.4 8080
Trying 172.19.1.4...
Connected to 172.19.1.4.
Escape character is '^]'.
Connection closed by foreign host.
root@webserver-s:/home/sushmi#
```

WE CAN BROWSE THE TOMACT FOR INTERNET.WE CAN ADD THE RULE MY IP ADDRESS



Not secure http://20.46.181.220:8080

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/11.0.18

THE ASF

If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

Security Considerations How-To

Manager Application How-To

Clustering/Session Replication How-To

Server Status

Manager App

Host Manager

Developer Quick Start

Tomcat Setup | Realms & AAA | Examples | Servlet Specifications

First Web Application | JDBC DataSources | | Tomcat Versions

Managing Tomcat

For security, access to the manager webapp is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 11.0 access to the manager application is split between different users. Read more...

Release Notes

Changelog

Migration Guide

Security Notices

Documentation

Tomcat 11.0 Documentation

Tomcat 11.0 Configuration

Tomcat Wiki

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

Tomcat 11.0 Bug Database

Tomcat 11.0 JavaDocs

Tomcat 11.0 Git Repository at GitHub

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

tomcat-announce
Important announcements, releases, security vulnerability notifications. (Low volume).

tomcat-users
User support and discussion

taglibs-user
User support and discussion for Apache Taglibs

tomcat-dev
Development mailing list, including commit messages